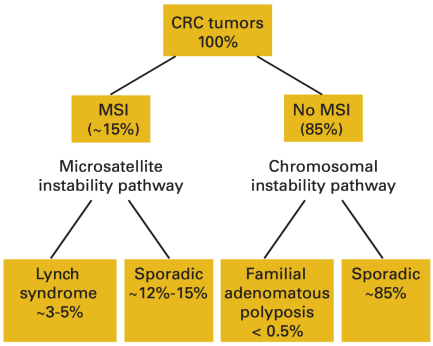
# Genetics in Colorectal Cancers

## Genetics of Colorectal Cancer




(Chapelle and Hampel 2010)

## Lynch Syndrome

- 3% of all colorectal cancers
- Germline mutations of mismatch repair genes
  - *OR* Germline deletions in EPCAM gene
- 80% lifetime risk colorectal cancer
- Early onset ($\tilde{x}$ 45-60yrs)
- Tumors tend to be right-sided

(Eng et al. 2022)

## Mismatch Repair Protein Loss in Lynch Syndrome

Lynch Syndrome can be caused by loss of expression of:

- MLH1
- PMS1
- MSH6
- MSH2
- MSH5

## Mismatch-repair Deficient Colorectal Cancers

Deficiency in mismatch repair proteins → microsatellite instability

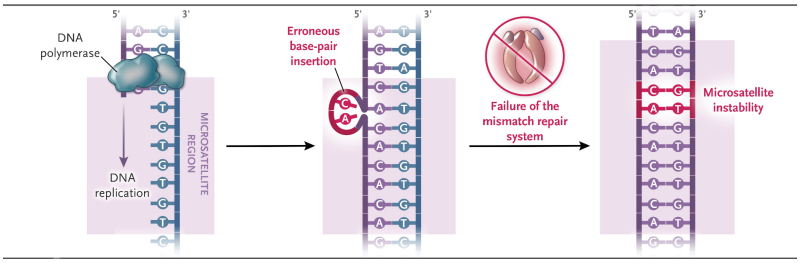
15% of colorectal cancers are MMR-deficient (MSI-High)

- Sporadic - Promoter methylation of MLH1 by BRAF V600E mutation (10%)
- Lynch - Due to mutations in mismatch repair genes (5%) 90% of Lynch tumors are MMR-deficient

*Reflex testing for BRAF V600E mutation is essential for MMR-deficient tumors*

## Mismatch Repair Proteins and Microsatellite Instability

Deficiency in mismatch repair proteins → microsatellite instability



## Mismatch Repair and MSI Testing

MMR Protein expression can be detected by immunohistochemistry

- Routine testing of tumor for GI cancers at CMC
- Loss of MLH1 leads to loss of PMS2

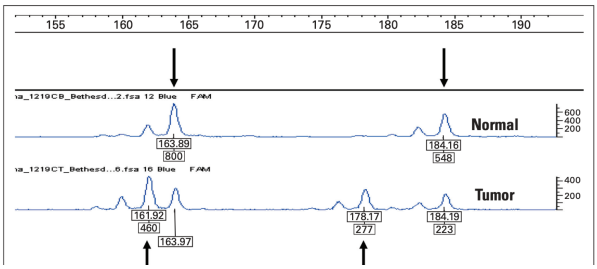
Microsatellite Instability can be detected by pCR

- Tumors are not routinely tested for MSI at CMC

97-99%+ concordance between dMMR and MSI-High status

## Mismatch Repair -> Microsatellite Instability

| Repeat Type | Sequence | Name and Location |
|---|---|---|
| Mononucleotide | AGGTAAAAAAAAAAAAAAAAAAAAAAAAAGGGT<br>$(A)_{26}$ shown | *BAT26*, Intron 5 of *MSH2* gene<br>Chromosome 2 |
| Dinucleotide | TGTACACACACACACATCGA<br>$(CA)_{6}$ shown | D5S346<br>Chromosome 5 |
| Tetranucleotide | ATATTCTATCTATCTATCTATCTATCTG<br>$(TCTA)_{5}$ shown | D14S608<br>Intergenic region chromosome 14 |



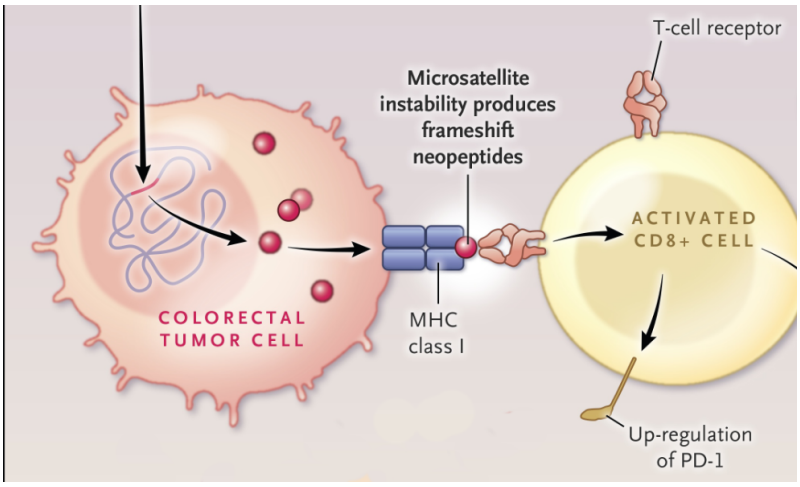

## MMR-deficient Tumor Clinical Feaures

- Proximal location
- Poor differentiation
- Mucinous histology
- Better prognosis (in early stage)
- Respond well to immunotherapy (ICI)
- Less responsive to 5-FU chemotherapy (in stage II)

(Taieb et al. 2022)

## MMR-deficient Tumors are Immunogenic

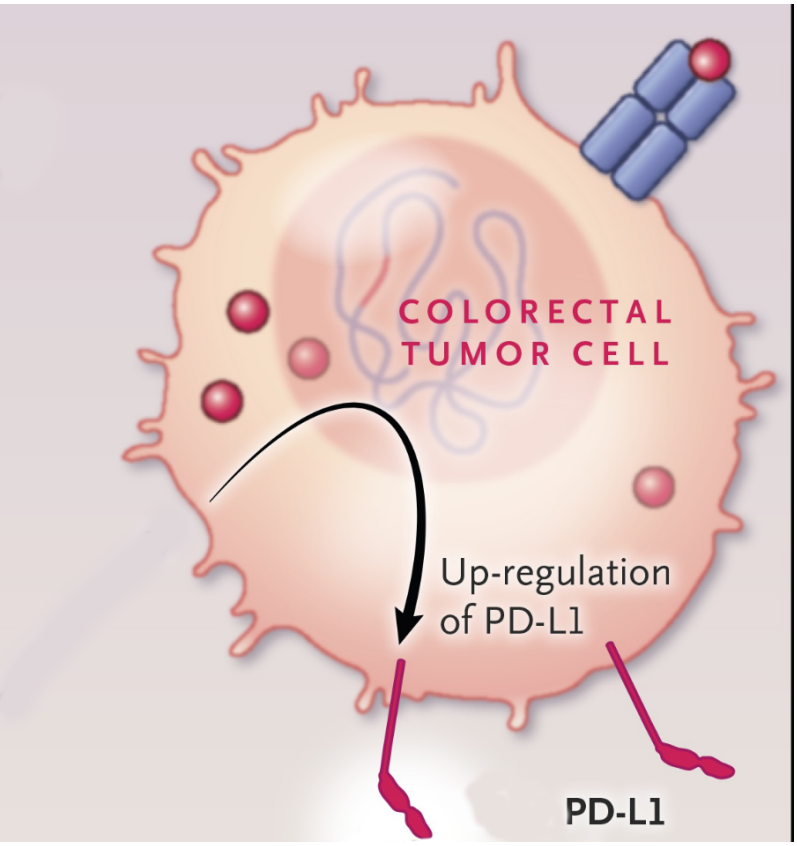
Mutational burden from dMMR causes MSI-High tumors to be immunogenic

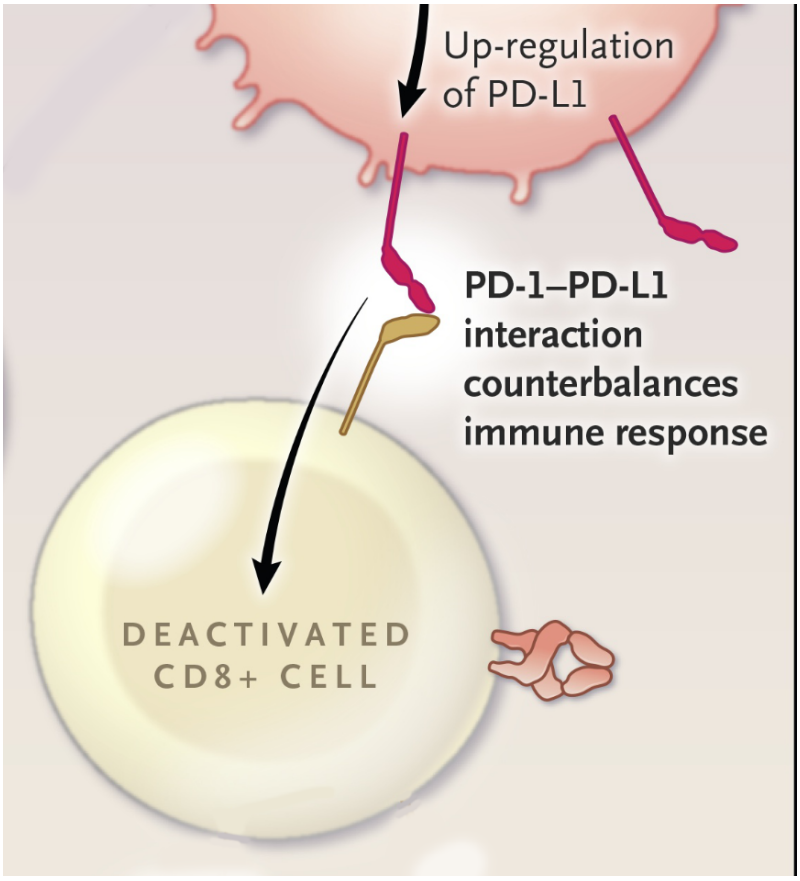



Activated tumor-infiltrating lymphocytes up-regulate PD-1

## Tumor PD-L1 Down-regulates Immune Responses

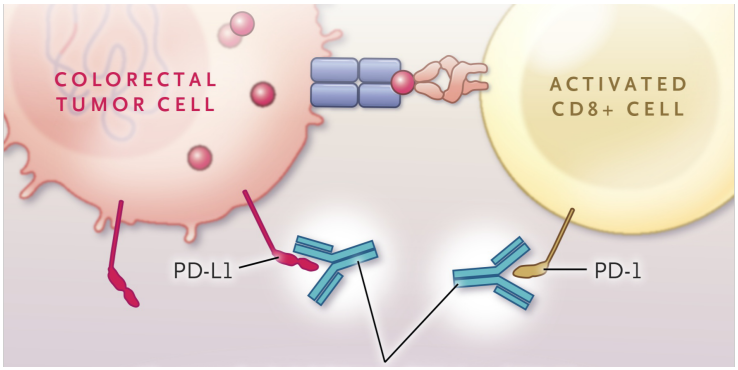
Tumor cells respond to immune activation by ↑ PD-L1


COLORECTAL
TUMOR CELL
Up-regulation
of PD-L1
PD-L1




↑ PD-L1 downregulates immune response

**Immune Checkpoint Inhibitors**



Inhibition of PD-L1 $\leftrightarrow$ PD-1 activates T cells by removing inhibition

## PD-L1 expression is higher in MMR-deficient CRC

Immunohistochemistry analysis of 1800 colorectal cancer specimens:

PD-L1 was more often positive in dMMR (18.6%) than in MMR proficient (pMMR) cancers (4.1%; $p < 0.0001$)

(Möller et al. 2021)

## MSI-High Colorectal Cancers

Clinical Relevance of Microsatellite Instability in Colorectal Cancer. Journal of Clinical Oncology : Official Journal of the American Society of Clinical Oncology. 2010. de la Chapelle A, Hampel H.

## BRAF V600E Colorectal Cancer Features

1. Age >70
2. Female
3. Proximal Right sided tumors
4. MLH1 hypermethylation
5. High grade and poor differentiation
6. More peritoneal and nodal metastasis
7. Less lung metastais

## Neoadjuvant Rx for MMR-deficient Colon Cancer - NICHE

115 colon cancer: neoadjuvant nivolumab + ipilimumab /*rightarrow* surgery

109 patients had a pathological response

105 Major pathological respoinse (TRG1 or TRG2)

- 75 pathological complete response

No recurrence of disease at median follow up of 26 months

(Chalabi et al. 2024)

## Adjuvant Therapy Stage II MMR-deficient Colon Cancer

ASCO: No benefit of treatment of stage II dMMR colon cancer with 5-FU alone

Ppooled analyses showing no difference in disease-free survival (HR 0.89, 95% CI 0.45-1.74) or overall survival (HR 1.03, 95% CI 0.53-2.02) with adjuvant chemotherapy in this population.

(Ribic et al. 2003)

## Adjuvant Therapy of Stage III MMR-deficient Colon Cancer

Oxaliplatin-based chemotherapy (FOLFOX or CapeOX) and atezolizumab (Tecentriq)

ATOMIC trial: atezolizumab + mFOLFOX6 is superior to mFOLFOX6 alone, with a 3-year disease-free survival rate of 86.4% versus 76.6% (hazard ratio 0.50, P .0001)

## FAP screening

Colonoscopy age 10-12 EGD for duodenal polyps at age 20-30 CT 1-3 years after colectomy and q5 yers in those with family hx of desmoids

## Lynch Non-Colorectal Cancers

- Endometrial (25-60% penetrance)
- Stomach
- Ovarian
- Urinary tract
- Biliary Tract
- Small bowel
- Male breast cancer (1.2% penetrance vs 0.1%)

(Sinicrope 2018)

## Lynch Screening

Colonoscopy q1-2 years staring age 20-25

## Colon Polyposis: >10 adenomatous polyps

- Classical FAP
- Attenuated FAP (AFAP)
- MUTYH-associated polyposis (MAP)
- Colonic adenomatous polyposis of unknown etiology (CPUE)

## Colon Polyposis: >4 hamartomatous polyps

- Puetz-Jaghers
- Juvenile Polyposis Syndrome
- Cowden Syndrome/PTEN Hamartoma Tumor Syndrome

## Serrated Colon Polyps

5 serrated polyps/lesions proximal to the rectum, all being 5 mm in size, with 2 being 10 mm in size OR >20 serrated polyps/lesions of any size distributed throughout the large bowel, with 5 being proximal to the rectum

## FAP

(FAP) is caused by germline mutations in the APC gene and leads to the development of hundreds to thousands of colorectal adenomas, with nearly 100% risk of colorectal cancer if untreated.

## Attenuated FAP

~30 polyps 70% penetrance by age 80 - mean age at dx 50-55

## References


Chalabi, Myriam, Yara L. Verschoor, Pedro Batista Tan, et al. 2024. "Neoadjuvant Immunotherapy in Locally Advanced Mismatch Repair-Deficient Colon Cancer." *The New England Journal of Medicine* 390 (21): 1949–58. https://doi.org/10.1056/NEJMoa2400634.

Chapelle, Albert de la, and Heather Hampel. 2010. "Clinical Relevance of Microsatellite Instability in Colorectal Cancer." *Journal of Clinical Oncology: Official Journal of the American Society of Clinical Oncology* 28 (20): 3380–87. https://doi.org/10.1200/JCO.2009.27.0652.

Eng, Cathy, Alexandre A. Jácome, Rajiv Agarwal, et al. 2022. "A Comprehensive Framework for Early-Onset Colorectal Cancer Research." *The Lancet. Oncology* 23 (3): e116–28. https://doi.org/10.1016/S1470-2045(21)00588-X.

Möller, Katharina, Niclas C. Blessin, Doris Höflmayer, et al. 2021. "High Density of Cytotoxic T-Lymphocytes Is Linked to Tumoral PD-L1 Expression Regardless of the Mismatch Repair Status in Colorectal Cancer." *Acta Oncologica (Stockholm, Sweden)* 60 (9): 1210–17. https://doi.org/10.1080/0284186X.2021.1933585.

Ribic, Christine M., Daniel J. Sargent, Malcolm J. Moore, et al. 2003. "Tumor Microsatellite-Instability Status as a Predictor

of Benefit from Fluorouracil-Based Adjuvant Chemotherapy for Colon Cancer." *The New England Journal of Medicine* 349 (3): 247–57. https://doi.org/10.1056/NEJMoa022289.

Sinicrope, Frank A. 2018. "Lynch Syndrome-Associated Colorectal Cancer." *The New England Journal of Medicine* 379 (8): 764–73. https://doi.org/10.1056/NEJMcp1714533.

Taieb, Julien, Magali Svrcek, Romain Cohen, Debora Basile, David Tougeron, and Jean-Marc Phelip. 2022. "Deficient Mismatch Repair/Microsatellite Unstable Colorectal Cancer: Diagnosis, Prognosis and Treatment." *European Journal of Cancer* (Oxford, England : 1990) 175 (November): 136–57. https://doi.org/10.1016/j.ejca.2022.07.020.